TEMPORAL DIFFERENCE LEARNING WITH SARSA

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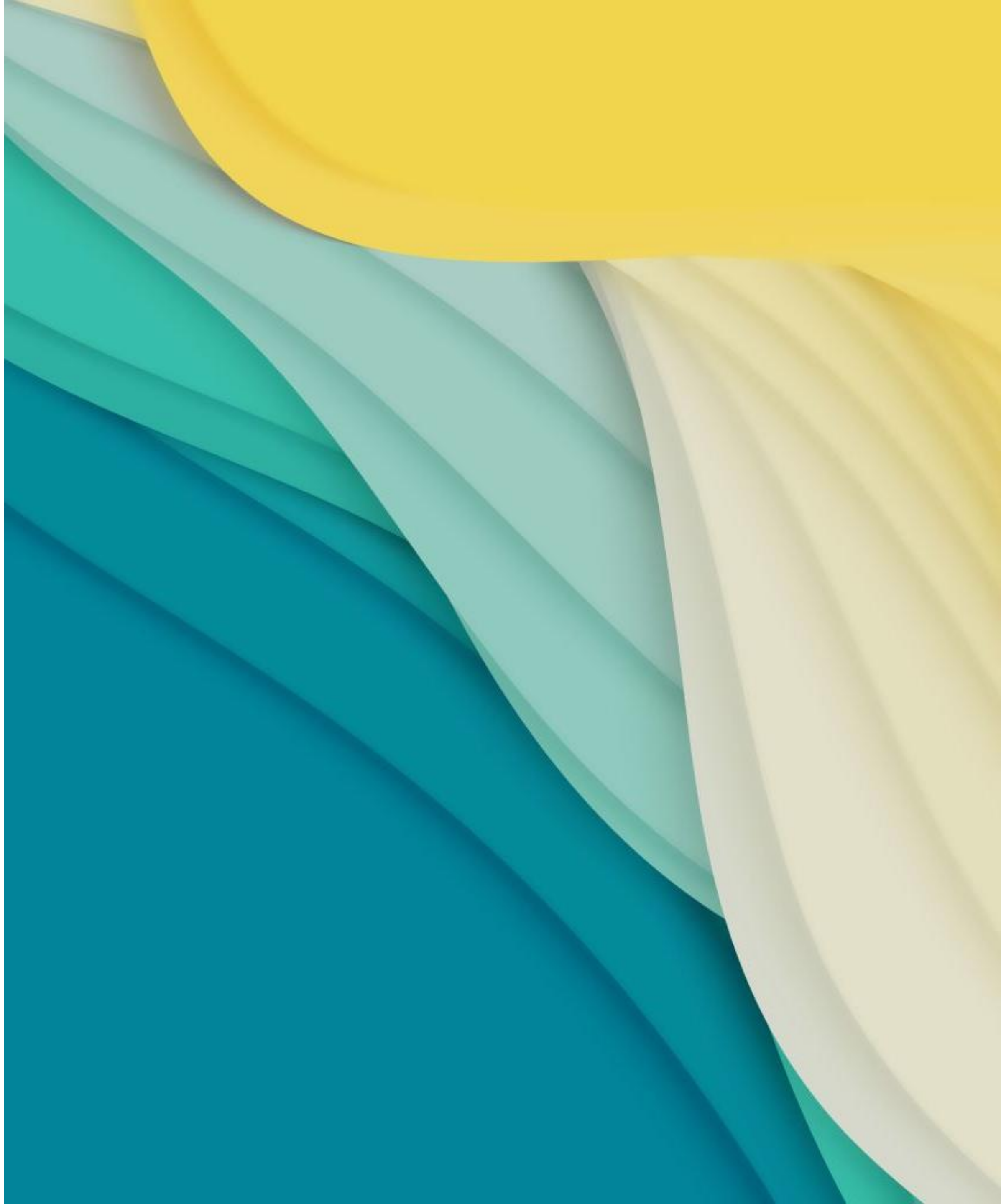

SARSA OVERVIEW

- SARSA stands for the sequence of events the agent experiences:
  - State → Action → Reward → Next State → Next Action
- It is a type of **reinforcement learning**
- Uses **on-policy** temporal difference control
- Learns while acting and updates every step using the actual next action it will take
- Uses agents, to learn by trial and error, to decide which actions to take so it can earn more rewards over time
- Process:
  - The agent observes the current state
  - It picks an action (ε-greedy)
    - Greed function mostly does the best-known thing, but will occasionally try new things to keep learning and improving
  - It gets a reward and goes to another state
  - From that next state, it picks the next action to do
  - Updates the model to decide, "how good was that first action in the first state?"
  - Repeat

REINFORCEMENT LEARNING

- Three types of reinforcement learning in chapter 19:
  - Dynamic learning
    - Relies on complete and accurate knowledge of environment dynamics
  - Monte Carlo (MC) based learning
    - Does not rely on complete and accurate knowledge of environment dynamics
    - Wait until the end of episode to calculate total return
  - **Temporal Difference (TD) Learning**
    - Do **not** have to wait until the end of episode to calculate total return (calculating as you go)
    - This is what SARSA is

TEMPORAL DIFFERENCE LEARNING

- Temporal Difference Learning can then be split into two main categories:
- **On-policy:**
  - Updates its values based on the actions the agent **actually** takes
  - In Plain English: You judge today's choice by looking at what you really did next, detours and all
  - **This is what SARSA is**
- Off-policy:
  - Updates its values using the highest possible reward from the next state no matter what the agent really did
  - Pretends you'll take the best possible next action, even if you didn't
  - Plain English: You judge today's choice by assuming that from the next step you'll make perfect choices
  - Q-Learning

Aspect	Q-learning	SARSA
Policy Type	Off-policy	On-policy
Update Rule	Uses the maximum Q-value from the next state	Uses the Q-value of the actual next action taken
Exploration	Encourages exploration by considering the best future action	Learns from the agent's actual policy and actions
Convergence Speed	Generally faster, as it assumes optimal actions	Slower, as it learns based on the actual actions taken
Stability	May lead to overestimation of Q-values	More stable, less prone to overestimation
Suitability	Suitable for environments where the optimal policy is the focus and aggressive exploration is acceptable	Suitable for environments where stability and alignment with the current policy are more important
Risk of Suboptimal Policy	Lower risk, as it always aims for the best possible action	Higher risk, as it learns based on the current policy which may be suboptimal

OVERVIEW/REAL WORLD EXAMPLE

- Temporal Difference Learning is a method that lets an agent **learn from experience** — **while it's still happening**, not just after an episode is finished.
- An example in chess:
    - You move a chess piece, then it looks at the board immediately and adjusts how good the position before was
    - That move that was done helped or hurt. So based off that information the model adjusts its rating of the last position up or down
- That's **temporal difference learning** — you learn from the **difference** between what you *expected* to happen and what did happen at the next time step.
- Where SARSA comes in, is the agent does what is most likely to be done vs. Q-learning assumes you do the best move
    - SARSA is more focused on policy
    - Q-learning is more focused on being aggressive and better for exploring

MATH- VARIABLES

s_t	state at time t
a_t	action taken at time t
r_{t+1}	reward received after taking action (a_t)
$\propto$	learning rate (between 0 and 1)
γ	discount factor (between 0 and 1). Tells how much you care about future rewards compared to immediate rewards.
$\mathbf{Q}(s, \mathbf{a})$	value estimate for taking action a in state s

MATH- GOAL

Goal: Find the optimal action-value function: $\boldsymbol{Q}^{*}(\mathbf{s}, \mathbf{a})$

That gives the expected long-term return if you

1. Start at state ***s***
2. Take action ***a***
3. Follow the best policy thereafter.

MATH- TEMPORAL DIFFERENCE TARGET

- The target is built from the next state and action you actually took. It’s a one step lookahead.

$$\boldsymbol{Target} = r_{t+1} + \gamma Q(s_{t+1}, a_{t+1})$$

MATH- TEMPORAL DIFFERENCE TARGET

$$\boldsymbol{Target} = r_{t+1} + \gamma Q(s_{t+1}, a_{t+1})$$

reward received after taking action (a_t)

How much we care about future rewards.
0 – only cares about immediate rewards.
1- values future rewards as much as present rewards

value estimate for taking the next action (a+1) in the next state (s+1)

MATH- THE UPDATE RULE

$$\boldsymbol{Q(s_t, a_t)} \leftarrow Q(s_t, a_t) + \alpha\Big[\underbrace{\big(r_{t+1} + \gamma Q(s_{t+1}, a_{t+1})\big)}_{\substack{\text{Target Function} \\ \text{(Estimate of true return)}}} - Q(s_t, a_t)\Big]$$

Value estimate for taking action *a* in state *s*

MATH- THE UPDATE RULE

$$\boldsymbol{Q}(\boldsymbol{s}_t, \boldsymbol{a}_t) \leftarrow Q(s_t, a_t) + \alpha\Big[\overbrace{\big(r_{t+1} + \gamma Q(s_{t+1}, a_{t+1})\big) - Q(s_t, a_t)}^{\textbf{TD Error}\text{ (how wrong our estimate was)}}\Big]$$

Value estimate
for taking action
a in state *s*

Learning Rate

MATH- THE UPDATE RULE

$$\boldsymbol{Q}(\boldsymbol{s}_t, \boldsymbol{a}_t) \leftarrow \underbrace{Q(s_t, a_t) + \alpha\left[\left(r_{t+1} + \gamma Q(s_{t+1}, a_{t+1})\right) - Q(s_t, a_t)\right]}$$

The whole thing is just a **small correction** toward a **better estimate**

-We stop when we converge, meaning the Q value stabilizes-

SOURCES

- https://www.geeksforgeeks.org/artificial-intelligence/differences-between-q-learning-and-sarsa/
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